



Problem Solving & Binary Tries

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Problem 1



Given a list of N strings ($a_1, a_2, \dots, a_N$) and Q queries. In each query you will be given a string X , you need to find whether there exist less than equal to 2 strings in the database such that the concatenation of them results in X .

$$1 \leq N \leq 10^3, \quad 1 \leq |a_i| \leq 10^2$$

$$1 \leq Q \leq 10^2, \quad 1 \leq |X| \leq 10^2$$

$a_1 = \underline{abc}$

$a_2 = \underline{ax}$

$a_3 = \underline{de}$

$a_4 = \underline{fg}$

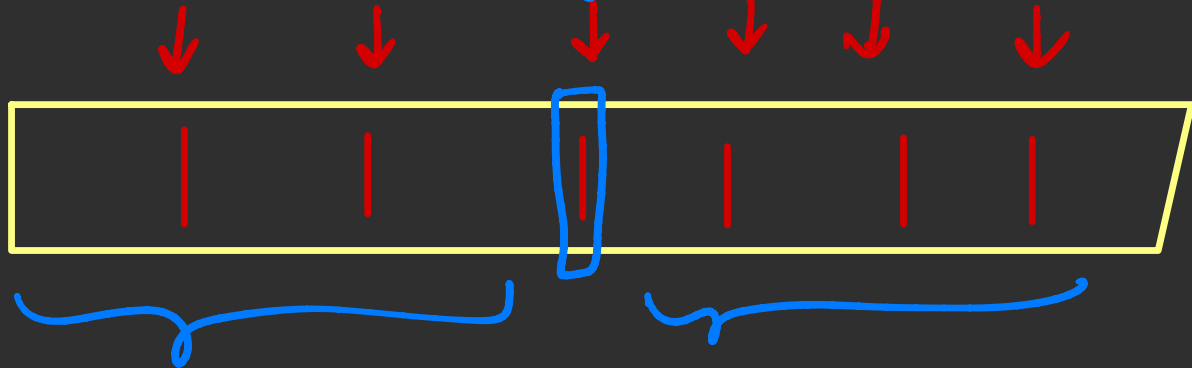
$X = abcde$ YES

$X = axax$ YES

$X = " "$ YES

$X = abe$ NO

$O(x) \cdot (2 \text{ hashing queries})$



$X =$

$\% \rightarrow 5 \text{ op}$

a | b | a | c | d | | a | a | f | g | h

Prefix

F

F

T

T

T

F

F

F

F

f

suffix

f

F

F

F

F

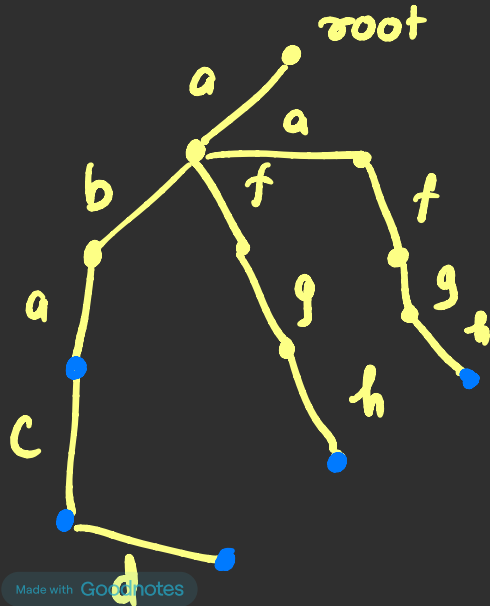
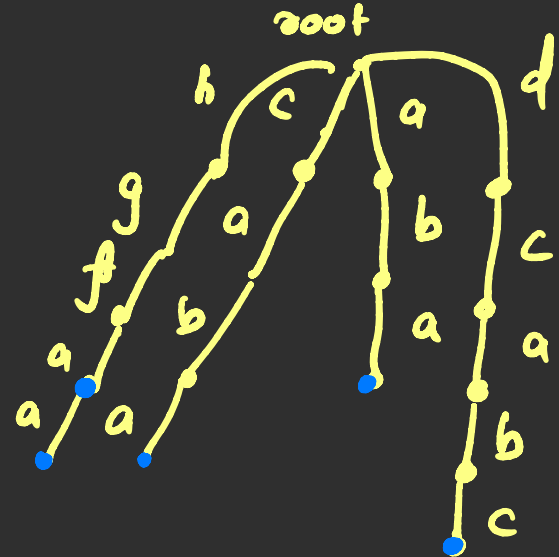
T

1

7

1

1


$$a_1 = abac$$
$$a_2 = aba$$
$$a_3 = abacd$$
$$a_4 = a/g h$$
$$Q_5 = aofgh$$
$$a_1' = caba$$
$$a_2' = aba$$
$$a_3' = d c a b a$$
$$a_4' = h g f a$$
$$a_s' = hgfaa$$


After we make 2 tries

Trie 1

On database

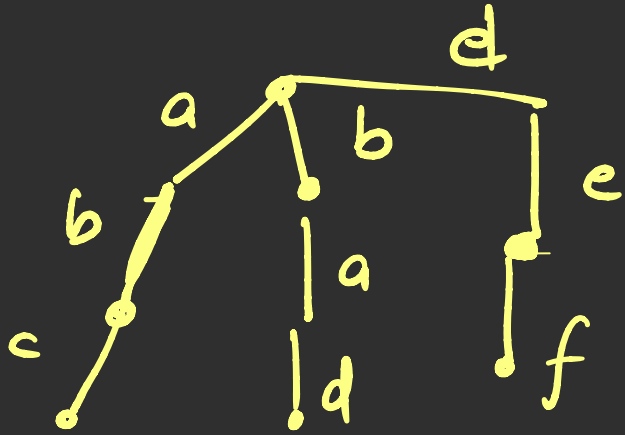
Trie 2

reversed
database

$$\text{pre}[i] = \text{suf}[i+1] = "T"$$

$$\underline{\underline{O(0.1 \times 1)}} \quad \underline{\underline{\Rightarrow TC}}$$

Space complexity of Trie



$N \rightarrow$ strings

$L \rightarrow$ length of each

$$O(N \cdot L \cdot 26)$$

$\uparrow$
26 nodes to each

$$\rightarrow N = 10^6, \quad L = 40, \quad C = 26$$

$$\begin{array}{ccccccc} 26 & 26 & 26 & 26 & 26 & \dots & 26 \\ \hline \hline \end{array}$$



All unique

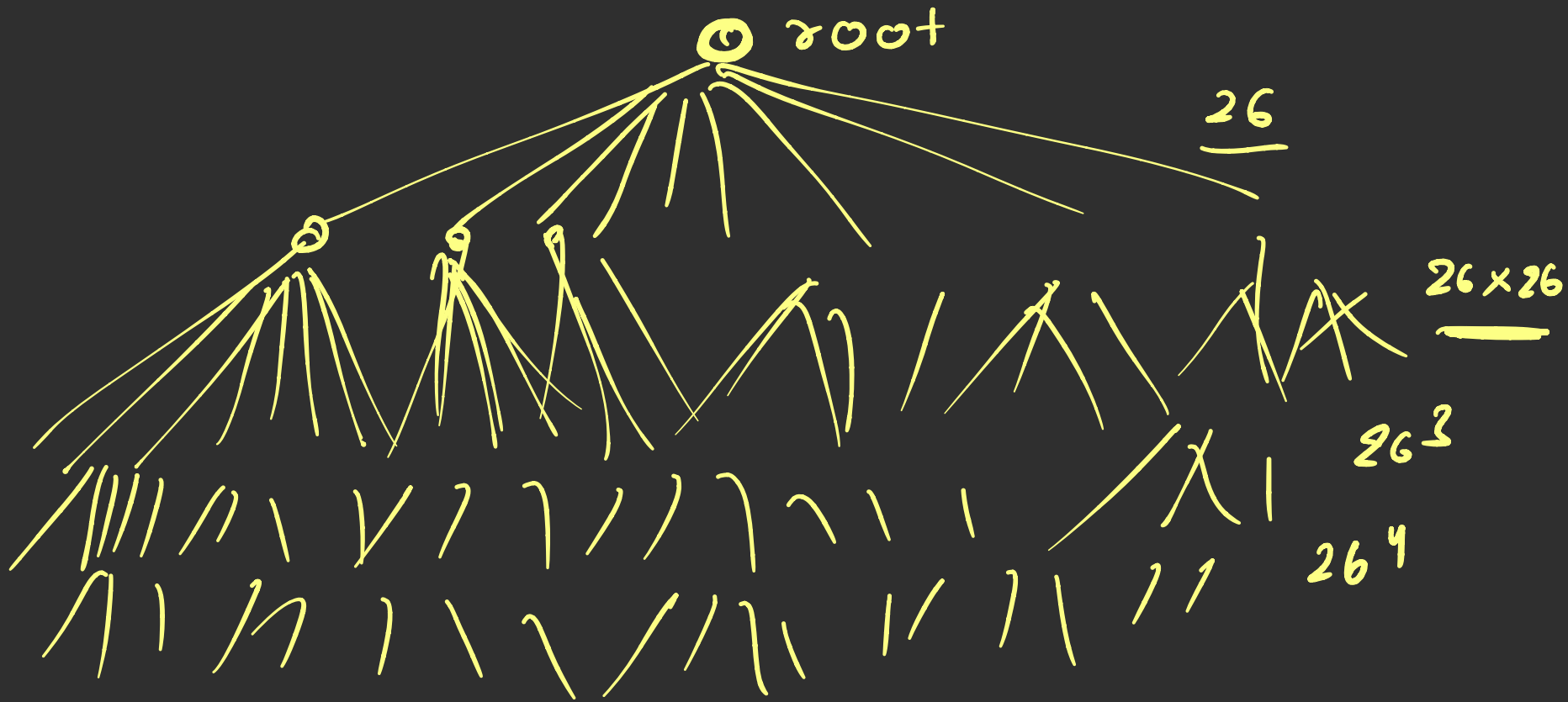
$$(26^{40}) = C^L$$

26⁴⁰

> > > > >

10⁶

N



Overestimate = N.L. 26

$$N = 10^6, \quad L = 40, \quad \underline{c} = 26$$

$$TN(10^6, 40, 26) =$$

$$26 + 26^2 + 26^3 + 26^4 + \dots + 26^{40}$$

$$\frac{26^x \leq N}{x \approx \log_{26} N} \rightarrow x \approx \log_c N$$

$$N = 10^6 \quad c = 26$$

$$\log_{26} 10^6 \approx \underline{\underline{4.24}} = 4$$

$$\hookrightarrow \frac{\log 10^6}{\log 26} \Rightarrow \frac{6}{\geq 1} \Rightarrow 4.24 = 4$$

$$26 + 26^2 + 26^3 \dots + 26^x \leq N$$

$$\left[26^{\downarrow} + 26^{\downarrow 2} + 26^{\downarrow 3} + 26^{\downarrow 4} + 10^6 \cdot 36^{\downarrow} \right]$$

$\Rightarrow$ TN (Branching Factor)

$$\left[26^2 + 26^3 + 26^4 + 26^5 + 10^6 \cdot 36 \cdot 26 \right]$$

N.L.26

$$\overset{A}{10^6 \cdot 40 \cdot 26} \mid \overset{B}{26^2 + 26^3 + 26^4 + 26^5 + \underline{10^6 \cdot 36 \cdot 26}}$$

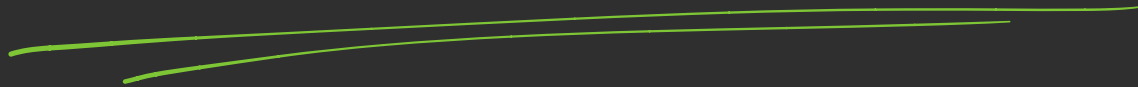
$$A > B \quad \checkmark$$

$$A < B$$

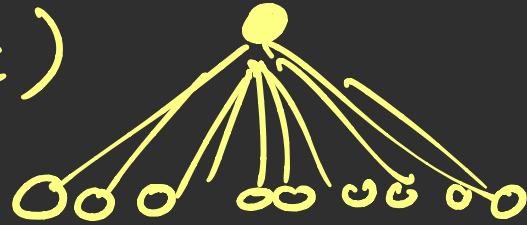
$$\underline{10^6 \cdot 4 \cdot 26} \mid \underline{26^2 + 26^3 + 26^4 + 26^5}$$

A is larger than B

by 8 times



Level 1
→ $26 + \text{TN}(N, L-1, C)$



Level 2
→ $26 + 26 * 26 + \text{TN}(N, L-2, C)$

Level 3
→
:
:
:
:
:



Binary Tries

Representing decimal numbers in binary, and insert their bits into a trie.

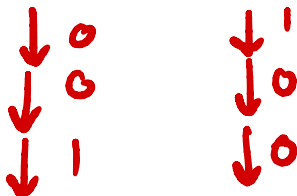
Trie Values: {0,1}

How to insert?

- MSB to LSB? ✓
- ~~LSB to MSB?~~ ✓
MSB

$$BF = 2$$

$$4 = (100)$$





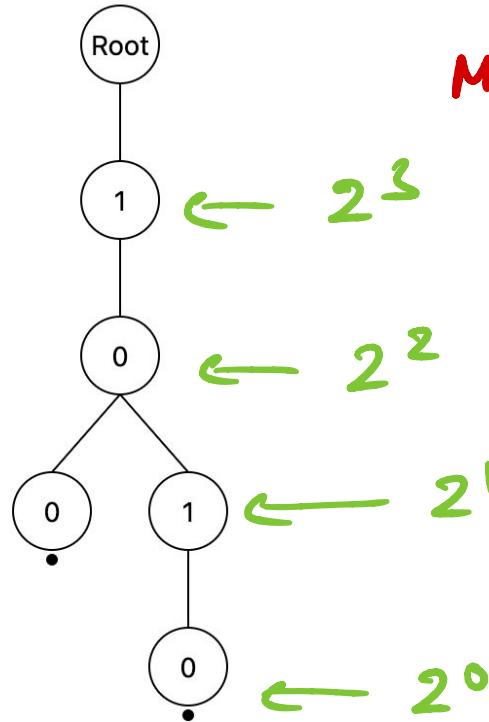
Binary Tries

Example:

$$A = \overset{\downarrow}{(1)}\overset{\downarrow}{(010)}_2 = 10$$

$$B = \overset{\downarrow}{(1)}\overset{\downarrow}{(00)}_2 = 4$$

Problem?



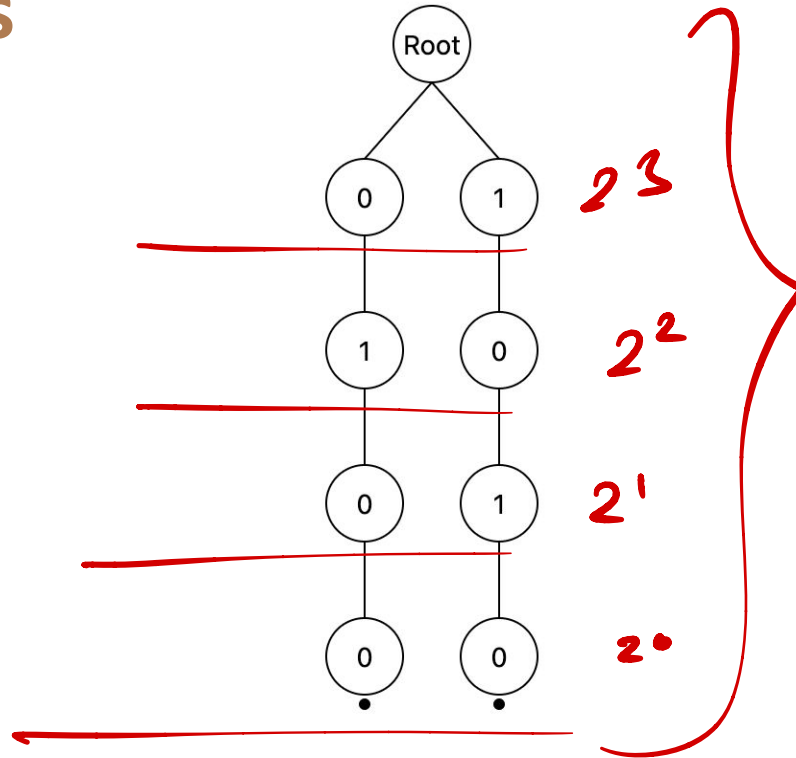


Binary Tries

Example:

$$A = (1010)_2 = 10$$

$$B = (\underline{0}100)_2 = 4$$



$$a_i \leq 10^9 \approx 32 \text{ bits}$$

$$a_i \leq 10^{18} \approx 64 \text{ bits}$$

$$4 = (100)_2$$

$$\begin{array}{ccccccc} 31 & & & & & & 0 \\ 0 & 0 & 0 & 0 & \dots & 29 & 100 & 0 \\ \hline & & & & & & & \end{array}$$



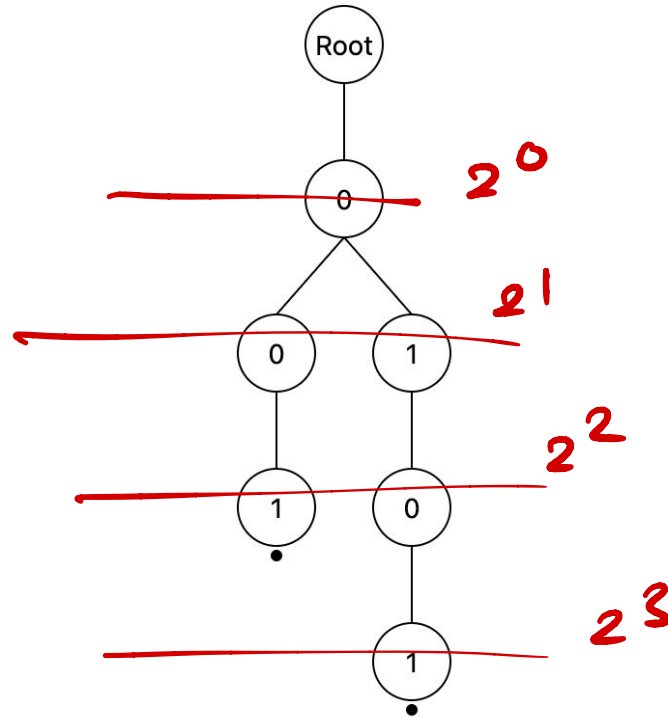
Binary Tries

Example:

$A = (1010)_2$

$B = (0100)_2$

Reverse?





Problem 2

Given an array A of N integers ($0 \leq a_i \leq 1e9$), find the maximum XOR of 2 numbers.

$$a[i] \oplus a[j] = \text{max} \uparrow$$

$$\times a_1 = 101010$$

$$\times a_2 = 000010$$

$$\checkmark a_3 = 001001$$

$$\times a_4 = 001000$$

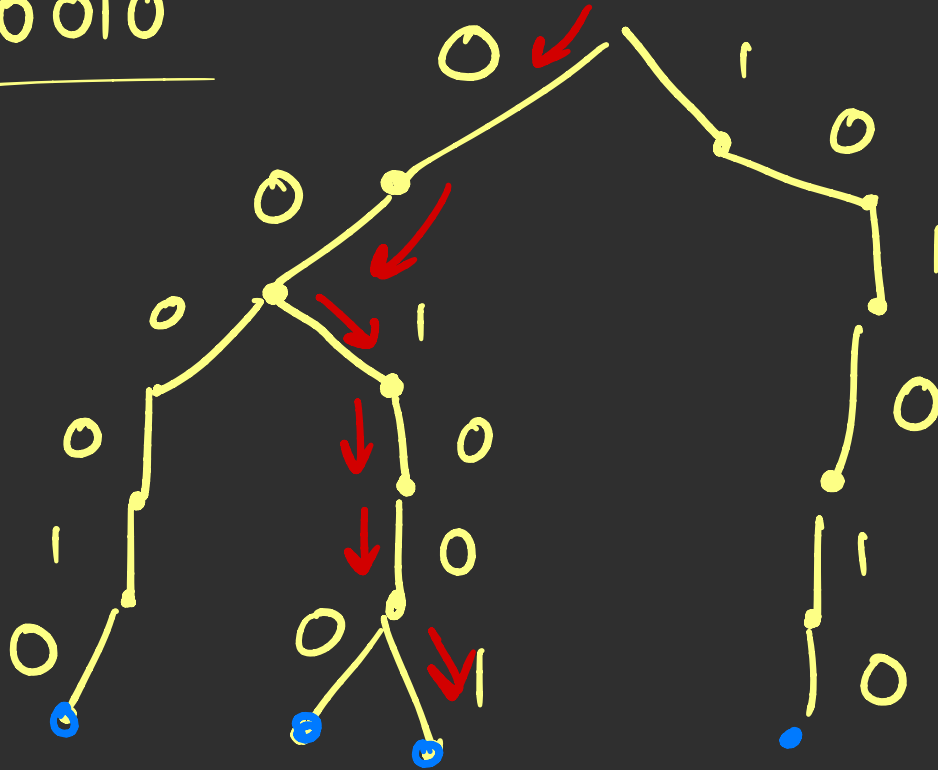
$$X = 110010$$

$$X \hat{=} a_3$$

Ans =

$$\begin{array}{cccccc} 1 & 1 & 1 & 0 & 1 & 1 \\ \hline \end{array}$$

$X = 110010$

root

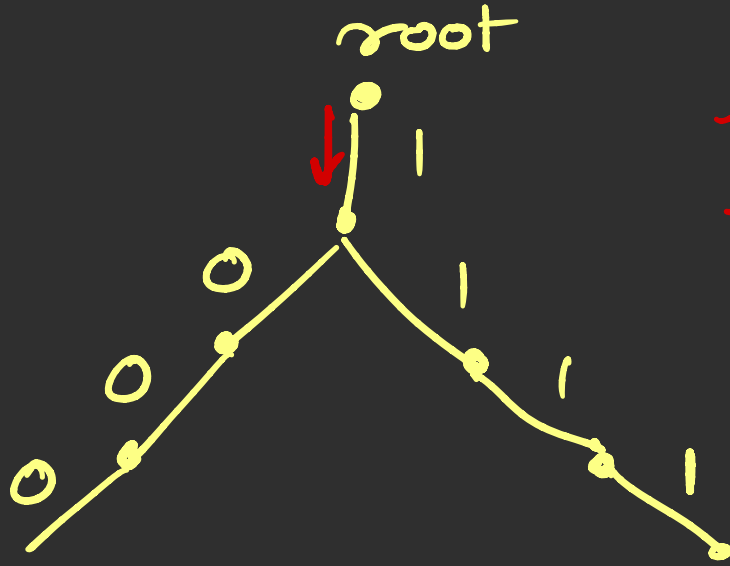


Problem 3

Given an array A of N integers ($0 \leq a_i \leq 1e9$), find the maximum AND of 2 numbers.

$X = 1001$

Scam!!!



Go left
Go right

X No deterministic factor

ans = 10001

01000
✓ 10011
✓ 10001
10000

00011

↓ ↓ ↓ ↓ ↓
1 0 0 0 1

≥ 2



Problem 4 [Homework]

Given an array `arr` of size `N` consisting of non-negative integers. You are also given `Q` queries represented by 2D integer array `queries`, where `queries[i] = [xi, mi]`.

The answer to the `i`th query is the maximum bitwise XOR value of `xi` and any element of `arr` that does not exceed `mi`. In other words, the answer is $\max(arr[j] \text{ XOR } xi)$ for all `j` such that `arr[j] ≤ mi`. If all elements in `nums` are larger than `mi`, then answer is -1.